

REVIEW

Pulse Field Ablation: From Bench to Blueprint: Mechanistic Insights and Device Engineering and Early Clinical Translation

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ABSTRACT: Pulsed field ablation has rapidly reshaped the landscape of atrial fibrillation ablation by offering a nonthermal, tissue-selective alternative to conventional thermal energy. This review highlights the scientific foundations, preclinical evidence, and engineering advances that established pulsed field ablation as a safer, faster, and more precise technology. We discuss how irreversible electroporation achieves myocardial selectivity, the pivotal studies that confirmed preservation of nearby structures, and the evolution of modern catheter systems designed for efficient and durable lesion creation. As pulsed field ablation platforms continue to mature, they are redefining procedural safety standards and expanding therapeutic possibilities in cardiac electrophysiology.



Key Words: atrial fibrillation ■ catheter ablation ■ echocardiography ■ electrophysiology ■ electroporation

Catheter ablation has become a cornerstone of rhythm control therapy for atrial fibrillation (AF) and is recommended by contemporary guidelines as a first-line strategy for maintaining sinus rhythm and improving quality of life in appropriately selected patients.^{1,2} Conventional thermal modalities, including radiofrequency ablation (RFA) and cryoballoon ablation (CBA), achieve pulmonary vein isolation (PVI) through the creation of continuous transmural lesions but remain associated with the risk of collateral injury to adjacent structures such as the esophagus, phrenic nerve, and pulmonary veins (PVs).^{3–5}

Pulsed field ablation (PFA) delivers ultrashort, high-voltage electrical pulses that induce irreversible electroporation of myocardial cells while largely preserving surrounding noncardiac tissues.^{6–8} This nonthermal, tissue-selective mechanism directly addresses one of the principal limitations of thermal ablation (TA).⁹ In this first of a 2-part review, we summarize the mechanistic foundations, preclinical evidence, engineering innovations, and early clinical translation that have established

PFA as a transformative advancement in catheter-based treatment of AF.

MECHANISM OF ELECTROPORATION

PFA generates intense, localized electric fields that exceed the critical threshold required to destabilize the phospholipid bilayer of the cell membrane.¹⁰ The resulting formation of nanoscale pores disrupts cellular homeostasis and leads to cell death through apoptosis or necrosis.¹¹ Although primarily a nonthermal process, secondary temperature elevations caused by Joule heating are inevitable but minimal. Thermal monitoring studies have shown that temperature rises during PFA rarely exceed 3°C at a tissue depth of 3 mm, which is insufficient to induce thermal necrosis and distinct from the injury mechanism seen with RFA.¹² Importantly, electroporation exists along a continuum between reversible and irreversible membrane permeabilization. At subthreshold electric field strengths or insufficient pulse dosing, transient nanopore formation allows restoration of cellular

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Nonstandard Abbreviations and Acronyms	
AF	atrial fibrillation
CBA	cryoballoon ablation
LAA	left atrial appendage
LAEO	left atrial appendage occlusion
PFA	pulsed field ablation
PFCA	pulsed field cryoablation
PV	pulmonary vein
PVI	pulmonary vein isolation
RFA	radiofrequency ablation
TA	thermal ablation

homeostasis, a phenomenon known as reversible electroporation. In contrast, when a critical electric field threshold is exceeded, irreversible electroporation results in sustained membrane disruption and cell death. The transition between these states is governed by field strength, pulse duration, pulse number, and tissue-specific electrical properties, establishing a dose–response relationship that directly determines lesion formation.¹³ Clinically, this distinction has important electrophysiological implications. Subthreshold energy delivery may result in transient conduction block despite preserved cellular viability, reflecting reversible electroporation rather than permanent lesion formation.¹⁴ This phenomenon underlies the concept of pulsed field–based functional mapping, in

which transient suppression of conduction can be used to identify arrhythmogenic substrates.¹⁵ However, it also highlights that acute electrogram elimination does not necessarily indicate durable tissue injury.

Cardiomyocytes display a greater susceptibility to irreversible electroporation than neighboring cell types such as vascular smooth muscle, endothelium, and neural tissue.^{16,17} This differential sensitivity is attributed to unique structural and electrophysiological properties of cardiac myocytes, including larger cell size, abundant transverse tubules, and specialized membrane architecture that favor electric field concentration.¹⁸ This preferential susceptibility is further supported by biophysical models of electroporation, which demonstrate that transmembrane potential induction is strongly influenced by cell size, geometry, and orientation relative to the applied electric field.^{19,20} Larger, elongated cells such as cardiomyocytes experience greater membrane polarization compared with smaller or more spherical cells, increasing the likelihood of exceeding the threshold for irreversible electroporation.²¹ In addition, the anisotropic organization of myocardial tissue, with aligned fiber orientation and dense transverse tubular networks, promotes localized electric field amplification and heterogeneous membrane charging.²² Variations in membrane composition and ion channel distribution may further modulate electroporation thresholds across cell types, collectively contributing to the preferential susceptibility of cardiomyocytes to irreversible electroporation. Histopathologic evaluations consistently demonstrate sharply demarcated myocardial

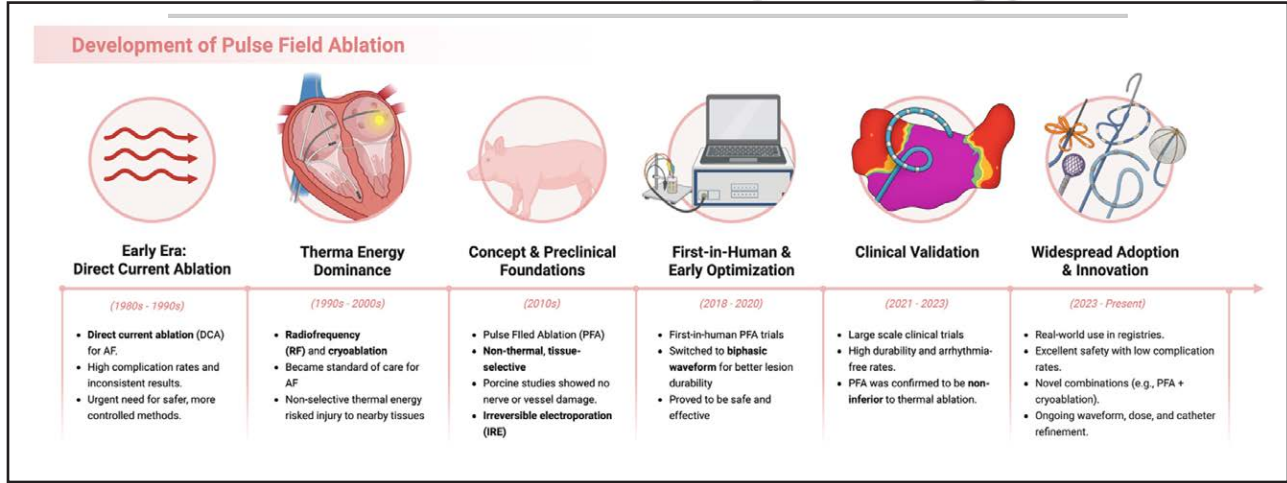


Figure 1. The historical development and clinical trajectory of pulsed field ablation (PFA). This timeline illustrates the evolution of energy sources for cardiac ablation, culminating in the current clinical use and innovation surrounding PFA. The initial era of direct current ablation (DCA) in the 1980s was characterized by inconsistent results and high complication rates. This led to the dominance of thermal energy methods, specifically radiofrequency (RF) and cryoablation, which became the standard of care in the 1990s and 2000s. The foundational concept and preclinical studies for PFA (involving tissue-selective, nonthermal irreversible electroporation [IRE]) were established in the 2010s, primarily using porcine models. The first-in-human and early optimization phase (2018–2020) saw the successful introduction of PFA using biphasic waveforms to improve lesion durability and proved the method safe and effective. PFA entered the clinical validation phase (2021–2023) through large-scale trials, confirming its excellent safety profile, durability, and noninferiority to thermal methods. The current era (2023–present) is focused on widespread adoption and innovation, including real-world use in registries, combination therapies (eg, PFA+cryoablation), and ongoing refinement of waveforms, dosing, and catheter technology. AF indicates atrial fibrillation.

necrosis with preservation of the extracellular matrix, vasculature, and adjacent noncardiac structures, reinforcing the concept of selective myocardial injury.^{1,23}

In addition to its structural selectivity, PFA also exerts functional effects on the cardiac autonomic nervous system, particularly at the level of atrial ganglionated plexi. Preclinical and early clinical observations suggest that PFA can induce transient neural stunning, leading to acute modulation of parasympathetic inputs that may contribute to arrhythmia suppression. Importantly, in contrast to thermal modalities, PFA appears to preserve neural structural integrity, consistent with its nonthermal mechanism. However, the extent and durability of autonomic modulation remain incompletely defined. Transient autonomic imbalance, including relative sympathetic predominance after parasympathetic withdrawal, has also been proposed as a potential mechanism underlying phenomena such as coronary vasospasm observed during PFA.

From a lesion formation standpoint, durability depends on achieving a uniform and complete transition to

irreversible electroporation across the target tissue. Heterogeneity in electric field distribution, influenced by tissue anisotropy, fibrosis, and impedance variability, may result in mixed zones of reversible and irreversible electroporation, predisposing to incomplete lesion formation and subsequent recovery of conduction.²⁴

Clinically, subthreshold energy delivery may induce reversible electroporation, resulting in transient electrical silence despite preservation of cellular viability, often described as stunned myocardium. These regions may demonstrate acute conduction block but subsequently recover, contributing to PV reconnection and variability in long-term lesion durability.²⁴ This phenomenon has also enabled the development of pulsed field–based reversible mapping, in which transient conduction suppression may be used to identify arrhythmogenic substrates; however, it underscores that acute electrogram elimination does not necessarily reflect permanent tissue injury.²⁵

Experimental models have played a central role in characterizing electroporation thresholds and lesion

Table 1. Key Biophysical Parameters of PFA and Their Clinical Implications



Parameter	Typical ranges/examples	Primary biophysical role	Clinical/practical implications
Voltage amplitude	1.5–2.5 kV for most microsecond PFA systems; higher fields needed for nanosecond platforms	Sets electric field strength; determines whether reversible vs irreversible electroporation is achieved	Too low→reversible electroporation, transient block and reconnection; too high→loss of selectivity, more skeletal muscle capture and arrhythmogenicity
Pulse duration	Microsecond (20–100 μ s) for contemporary PFA; nanosecond for CellFX nano-PFA	Shorter duration limits conductive heating while still reaching transmembrane potential threshold	Microsecond pulses balance lesion depth and nonthermal profile; ns pulses minimize Joule heating further and may favor apoptotic pathways, but require higher peak fields
Pulse number/train structure	4–10 pulses per packet, repeated in 4–10 trains per application depending on system	Cumulative electroporation dose; increases lesion depth and homogeneity	More pulses improve transmural, especially in thicker tissue, but may add procedure time and small incremental heating; under-dosing predisposes to PV reconnection
Waveform polarity (monophasic vs biphasic)	Early monophasic; contemporary systems almost universally biphasic microsecond pulses	Charge balancing, reduction in net DC offset; modulates membrane charging dynamics and skeletal muscle capture	Biphasic waveforms reduce skeletal muscle contraction, microbubble formation, and improve lesion homogeneity vs monophasic; now standard in clinical systems
Frequency/pulse repetition rate	Tens to hundreds of Hz for many microsecond systems; kHz repetition for some nano-PFA platforms	Governs temporal summation of electroporation and potential interaction with refractoriness	Higher repetition rates can shorten delivery time and augment lesion depth but may increase neuromuscular stimulation; optimization differs by platform
Bipolar vs monopolar field configuration	FARAPULSE, VARIPULSE and most lattice/grid catheters use bipolar fields; Centauri, Sphere-360, and some dual-energy systems use monopolar or mixed modes	Determines field geometry and depth; monopolar allows broader, deeper fields; bipolar confines field near electrodes	Bipolar reduces far-field exposure and muscle capture but may require closer contact; monopolar facilitates deeper lesions and linear work but with higher risk of off-target exposure if dosing is not tuned
Cardiac gating (synchronization to R-wave)	Most LA systems R-wave gated; some grid/niche technologies may use nongated trains	Reduces risk of induction of ventricular arrhythmias by avoiding vulnerable periods	Nongated delivery raises theoretical proarrhythmic risk; R-wave gating is standard in commercially approved PV systems
Contact dependence	Less contact-sensitive than RFA but still influenced by CF 15–30 g and electrode–tissue proximity	Ensures adequate field coupling and reduces current dispersion	Poor apposition leads to mixed reversible/irreversible zones and recovery; impedance and signal-based surrogates, plus CF on focal systems, help standardize delivery
Thermal contribution (Joule heating)	Usually <3°C at 3 mm depth in optimized protocols	Secondary effect of current flow; not primary mechanism	Excessive local heating (eg, high-voltage, prolonged trains, low flow) may introduce hybrid lesions and hemolysis; biomarker elevations vary by catheter footprint

Summary of voltage, pulse, waveform, and delivery parameters across contemporary microsecond and nanosecond pulsed field ablation platforms. Values represent typical ranges reported across contemporary clinical and preclinical PFA platforms; specific parameters vary by manufacturer and protocol. CF indicates contact force; DC, direct current; LA, left atrium; PFA, pulsed field ablation; PV, pulmonary vein; and RFA, radiofrequency ablation.

Adapted from Koruth et al 2019, Reddy et al,⁶⁹ García-Sánchez et al,²⁶ and Nakagawa et al 2024.

formation, including ex vivo tissue preparations and simplified substrates such as potato models, which provide a visually accessible surrogate for electric field distribution and lesion geometry. However, these systems have important limitations. Unlike myocardial tissue, they lack structural anisotropy, heterogeneous conductivity, and complex cellular architecture, all of which critically influence electric field propagation and electroporation thresholds. As a result, findings from such models may not fully translate to in vivo cardiac tissue, particularly with respect to lesion depth, field distribution, and tissue selectivity. These limitations underscore the need for careful interpretation of preclinical data and validation in biologically relevant models.

WAVEFORM CHARACTERISTICS AND ENERGY DELIVERY PARAMETERS

The efficacy and safety of PFA are governed by key energy delivery parameters, including voltage amplitude, pulse duration, pulse number, waveform configuration, and frequency.⁹ Voltage amplitude is the primary determinant of electric field strength and dictates whether the threshold for irreversible electroporation is achieved. Lower amplitudes may result in reversible electroporation and transient conduction block, whereas higher amplitudes produce permanent membrane disruption and durable lesion formation.^{26–29}

Pulse duration and pulse number further influence lesion depth and homogeneity. Shorter pulse durations, typically in the microsecond range, minimize conductive heating and preserve the nonthermal mechanism of action while increasing pulse number enhances cumulative electroporation effects and lesion contiguity. Repeated pulse delivery has been shown to augment lesion depth, particularly in thicker myocardial regions.^{17,18,24,26}

Waveform configuration represents a critical determinant of both safety and efficacy. Early monophasic waveforms were associated with greater skeletal muscle activation, bubble formation, and less uniform lesion geometry. In contrast, biphasic waveforms have become the standard in contemporary PFA systems due to their ability to reduce skeletal muscle activation and improve lesion homogeneity through charge balancing and more uniform electric field distribution.^{26,29,30}

Although PFA is primarily nonthermal, Joule heating may occur at higher energy settings or with prolonged pulse delivery. Temperature increases are generally modest but may contribute to localized thermal effects under certain conditions.^{31–33}

However, variability in how waveform parameters are defined and reported across platforms remains a significant limitation. Differences in voltage scaling, pulse definitions, and dosing strategies continue to limit cross-study comparability and highlight the need for standardized reporting.^{30,34}

HISTORICAL PRECEDENCE

The modern era of cardiac ablation began in the early 1980s when Melvin Scheinman performed the first percutaneous procedure using direct current energy.³⁵ Although effective, these high-energy shocks often caused barotrauma and unpredictable tissue injury.^{36,37} By the early 1990s, RFA replaced direct current as the preferred approach, producing controlled thermal injury through resistive heating and allowing real-time monitoring of lesion formation.³⁸ Its safety, precision, and reproducibility established it as the dominant technique for arrhythmia treatment.³⁹ Cryoablation, adapted from cryotherapy developed for tumor and frostbite management, preserved tissue structure and reduced the risk of perforation, but early devices were less practical and slower to use.⁴⁰

Electroporation-based ablation was initially explored in parallel with the evolution of thermal technologies, primarily in surgical and experimental settings. Early work by Lavee et al⁴¹ demonstrated the feasibility of irreversible electroporation as a nonthermal energy source for epicardial atrial ablation, achieving effective myocardial injury while preserving surrounding structures. Subsequent studies by Neven et al⁴² further characterized lesion formation using epicardial linear electroporation, highlighting the influence of electrode configuration and field distribution on lesion size and depth. Conceptual work by DeSimone and colleagues framed electroporation as a potential paradigm shift in catheter ablation, distinguishing it from both traditional direct current shock and thermal energy modalities.⁴³ In parallel, the Iowa Approach revisited direct current–based ablation strategies with a focus on controlled electroporation, particularly for applications near coronary arteries where thermal injury carries significant risk. Although these early systems were limited by imprecise energy delivery and lack of catheter-based integration, they established the mechanistic and safety foundations that would later enable the development of modern PFA technologies.⁴⁴

Building on these early electroporation studies, PFA later emerged as a catheter-based, nonthermal modality capable of selectively targeting myocardial tissue while sparing adjacent structures. Although early systems lacked the precision for clinical translation, these initial discoveries laid the scientific foundation for the preclinical development that would ultimately define modern PFA.^{30,45}

PRECLINICAL FOUNDATIONS

Early preclinical studies, largely conducted in porcine models, were essential in defining the 3 core principles of PFA: tissue selectivity, a nonthermal mechanism of action, and a strong dependence on tissue proximity.^{1,5,6,11} The selective effect of PFA was demonstrated through

Table 2. Contemporary PFA Catheter Systems and Platforms

Category	System/ company	Catheter architecture	Waveform/field configuration	Mapping/ integration	Key clinical data and status
Circumferential and large-area systems					
	FARAWAVE (±FARAFLEX/ FARAPOINT), Boston Scientific	Pentaspine, 20-electrode multispline flower catheter for antral PVI; FARAFLEX/ FARAPOINT provide mapping and focal lesions	Bipolar, biphasic microsecond pulses; R-wave gated; generator shared across FARAWAVE and FARAPOINT	Stand-alone generator; integration with Rhythmia HDx in newer workflows	ADVENT RCT: PFA noninferior to RF/Cryo for composite efficacy end point with similar atypical flutter incidence; early series show 96% PVI durability with optimized biphasic waveform; MANIFEST-PF (n≈1500) and MANIFEST-17K (n>17 000) show 99% acute PVI with no PV stenosis, atrioesophageal fistula, or permanent phrenic injury
	PulseSelect,	9-electrode circular PVI catheter	Biphasic pulses (described as monopolar delivery in early reports); microsecond duration; R-wave gated	generator; can be used alongside conventional mapping systems and within the Affera ecosystem	PULSED AF pivotal: 100% acute PVI; 0.7% primary safety end point; ≈73% to 80% 1-y arrhythmia freedom depending on AF type; favorable CIED-EMI profile in bench and clinical testing
	Volt Balloon, Abbott	Balloon-in-basket PFA catheter integrated with EnSite X; designed for single-shot antral PVI	Biphasic microsecond pulses in bipolar configuration	Tightly integrated with EnSite X 3-dimensional mapping	VOLT CE Mark study (n=146): 99.1% acute PVI; 2.7% 7-d safety events; VOLT-AF IDE (≈400 pts): 12-mo arrhythmia-free 83.5% paroxysmal and 58.1% persistent AF with low major complication rates; FDA approval 2025
	Sphere-360 (Affera System)	Large 30-mm spherical lattice catheter with multiple electrodes for single-shot PVI plus mapping	Monopolar biphasic microsecond pulses; R-wave gated; stepwise optimization in PULSE1–3	Fully integrated with the Affera Mapping and Ablation System	PULSE program: 98% per-vein PVI durability; 88% 1-y freedom from atrial arrhythmias in paroxysmal AF; mean procedure time ≈58 min and PVI in ≈12 min; excellent safety with no primary end point events
	Globe Mapping and Ablation System, Kardia	30-mm spherical basket with 122 electrodes enabling simultaneous high-density mapping and ablation	Bipolar, biphasic microsecond pulses	Stand-alone integrated mapping-ablation platform	PULSE-EU and PULSAR IDE: 1-y effectiveness ≈78% with no device- related primary safety events; allows rapid map-and-ablate workflows
Focal and intermediate-footprint systems					
	VARIPULSE+OMNYPULSE, Biosense Webster	VARIPULSE: variable- loop circular PFA catheter; OMNYPULSE: spherical-tip catheter for focal/linear work and high-density mapping	Bipolar, biphasic microsecond pulses	Fully integrated with CARTO 3; supports near-fluoro-less workflows	insPIRE and admIRE: 100% acute PVI, 12-mo arrhythmia-free ≈79% and 75% respectively; ≈25% of cases fluoroscopy-free; VARIPULSE FDA- approved Nov 2024; OMNYPULSE under evaluation in Omny-IRE trial
	FARAPOINT (±FARAFLEX), Boston Scientific	FARAPOINT: irrigated focal PFA catheter; used for CTI, lines, non-PV triggers, and touch-up after FARAWAVE	Shares FARAPULSE generator; biphasic microsecond pulses; bipolar field	Integrated within FARAPULSE system; can be combined with 3-dimensional mapping (Rhythmia/other)	ADVANTAGE-AF phase 2: FARAPOINT CTI ablation in 141 persistent AF pts yielded 96.4% flutter- free survival at 12 mo with favorable safety
	Sphere-9,	9-mm spherical lattice- tip catheter with 9 electrodes for focal PFA and RF	Dual-energy platform: biphasic microsecond PFA plus irrigated RF; various bipolar/monopolar configurations	Integrated with Affera Mapping and Ablation System	SPHERE Per-AF IDE: noninferior to conventional RF for persistent AF; primary effectiveness 73.8% vs 65.8% (<i>P</i> <0.0001 for noninferiority); enables PVI plus linear lesions from a single catheter
Emerging and niche technologies					
	CellFX nano-PFA, Pulse Biosciences	360° circular catheter (and other prototypes) connected to nano-PFA generator	Bipolar nanosecond pulses at kHz repetition; highly nonthermal, targeting intracellular structures	Stand-alone platform; mapping integration under development	First-in-human: 5 pts with complete PVI and no complications; 30-pt feasibility: 100% acute success, durable isolation at 28 d, transient phrenic stunning and mild esophageal changes resolving by 2–4 wk; multicenter IDE trial registered 2024 (URL: https://www.clinicaltrials.gov ; Unique identifier: NCT06696170)

(Continued)

Table 2. Continued

Category	System/ company	Catheter architecture	Waveform/field configuration	Mapping/ integration	Key clinical data and status
iCLAS PFCA, Adagio Medical		Flexible catheter capable of ultralow-temperature cryo and PFA; designed for wide lesions (PVI, PWI, CTI)	PFCA: precooling plus biphasic microsecond pulses, minimizing bubbles and muscle contraction	Adagio mapping tools; compatible with standard EP workflows	PARALELL FIH (66 persistent AF pts): high acute PVI/PWI success ($\approx 96\%$ – 98%), low serious events (1 groin hematoma, 1 intubation-related complication); long-term durability under evaluation
Centauri System, CardioFocus		Uses standard CF-sensing focal RF catheters (TactiCath SE, IntellaNav StablePoint, ThermoCool ST) to deliver PFA via dispersive patch	Biphasic monopolar microsecond waveform (WAVE1) creating broader, deeper electric field	Leverages existing CARTO, EnSite, and Rhythmia mapping systems	ECLIPSE-AF: 100% acute PVI; 89% vein durability at 3 mo; no phrenic or esophageal injury; minimal vascular complications—demonstrates feasibility of focal CF-guided PFA with existing hardware
Nanopulse Platform, Pulse Biosciences		Catheter prototypes for bipolar nanosecond delivery (distinct from microsecond CellFX systems)	High-repetition-frequency nanosecond pulses; strongly nonthermal; higher peak fields required	Early development; mapping integration not yet standardized	Preclinical work suggests transmural lesions with minimal thermal rise and little skeletal muscle activation; human AF data still limited
Argamed platform, Arga MedTech		Catheter system with adjustable multielectrode designs (details evolving)	Proprietary biphasic or asymmetrical biphasic microsecond pulses with tunable voltage, pulse width, and frequency	Platform designed to interface with standard EP labs; mapping partners evolving	Early feasibility data indicate effective lesion formation with reduced neuromuscular stimulation; robust clinical AF data not yet available
Field Medical System		Catheter-based bipolar PFA system with proprietary preset pulse trains	Microsecond biphasic pulses; bipolar configuration with fixed dosing protocols to enhance reproducibility	Stand-alone system; designed to minimize off-target exposure and neuromuscular activation	Early FIH and FIELD-PULSE trial data presented in 2026 suggest high acute PVI success and favorable safety, but peer-reviewed publications are pending
Volt PFA Grid/Lattice, Abbott		Multielectrode grid/lattice catheter delivering wide-area PFA lesions	Microsecond biphasic pulses across distributed electrode array; not R-wave gated; high-current trains over several seconds	Integrated with EnSite X; complements Volt Balloon for WACA/linear work	Early clinical experience (European centers and CE-Mark program) shows efficient wide-area ablation with good lesion continuity and manageable muscle activation; detailed outcomes still limited

Categorized by ablation footprint: circumferential/large-area, focal/intermediate, and emerging/niche technologies. AF indicates atrial fibrillation; CF, contact force; CTI, cavotricuspid isthmus; EP, electrophysiology; FDA, Food and Drug Administration; FIH, first-in-human; IDE, Investigational Device Exemption; PFA, pulsed field ablation; PFCA, pulsed field cryoablation; PV, pulmonary vein; PVI, pulmonary vein isolation; PWI, posterior wall isolation; RF, radiofrequency; and WACA, wide-area circumferential ablation.

direct comparisons with RFA. In animal models, ablations performed near the esophagus caused injury in all subjects treated with radiofrequency energy, including cases of fistula formation and deep ulceration, while no injury occurred in those treated with PFA.⁴⁶ Similar findings were observed for the phrenic nerve, where careful mapping showed a dose-dependent but reversible response during energy delivery, and no structural damage was found on histological examination several weeks later. Additional studies confirmed that PFA preserves vascular integrity, as PVs treated with PFA showed minimal dimensional change over time compared with the significant narrowing seen after RFA. Collectively, these preclinical investigations established the essential biological and engineering principles that enabled the safe translation of PFA to human studies.⁴⁷ They demonstrated that the effectiveness of lesion formation depends strongly on electrode–tissue proximity and field uniformity and that appropriate energy dosing can achieve consistent transmural lesions while preserving vascular and neural

structures.^{48–50} These insights provided the technical blueprint that guided early human feasibility trials and informed subsequent catheter design and waveform optimization.⁵¹

FIRST-IN-HUMAN EXPERIENCE

In 2018, the first-in-human study of PFA for AF treated 22 patients using endocardial and epicardial approaches; acute PVI was achieved in all patients and surgical box lesions in 86%, with no acute adverse events. This pioneering work, using a pentaspline multielectrode catheter with monophasic waveforms, paved the way for rapid technological refinement. A key advancement was the transition to biphasic energy delivery, which was shown to reduce skeletal muscle engagement and significantly improve the durability of PVI.⁵²

The efficacy of this optimized approach was validated in the IMPULSE, PEFCAT, and PEFCAT II studies, which collectively enrolled 121 patients with paroxysmal AF.

These trials reported 100% acute PVI success and a 96% durability rate in patients receiving the optimized biphasic waveform.⁴⁴ At a median follow-up of 49 months, 73% of patients remained free from atrial arrhythmias.⁵³ Lastly, the ADVENT trial (N=607) established noninferiority of PFA to the TA (73.3% versus 71.3% freedom from composite end point at 12 months). Notably, the incidence of postablation atypical atrial flutter was similar between groups (5.2% versus 5.6%; $P=0.86$), addressing early concerns about the arrhythmogenic potential of PFA lesions.^{53,54}

REAL-WORLD EVIDENCE AND SAFETY PROFILE

Large, real-world registry data have played a central role in defining the safety and clinical uptake of PFA. In the MANIFEST-PF registry, which enrolled 1568 patients across 24 European centers, acute PVI was achieved in 99.2% of cases. At 1 year, 78.1% of patients remained free from AF, atrial flutter, or atrial tachycardia, with higher arrhythmia-free survival in those with paroxysmal AF (81.6%) compared with persistent AF (71.5%).⁵⁵

The subsequent MANIFEST-17K registry, encompassing 17 642 patients and representing the largest PFA safety data set to date, demonstrated low rates of major adverse events. Notably, there were no reported cases of atriopharyngeal fistula, PV stenosis, or phrenic nerve palsy. Other complications were uncommon, including pericardial tamponade (0.36%), stroke (0.12%), and coronary artery spasm (0.14%).⁵⁶

These findings reinforce the favorable safety profile observed in preclinical studies and support the rapid clinical adoption of PFA.

LESION CHARACTERIZATION

Lesion characterization in PFA is defined by its nonthermal mechanism, resulting in selective myocardial injury with preservation of the extracellular matrix and surrounding structures.

Histopathologic evaluation of PFA lesions demonstrates sharply demarcated zones of myocardial injury with preservation of the extracellular matrix, microvasculature, and adjacent noncardiac structures. In the acute phase, affected myocardium exhibits cellular swelling, contraction band necrosis, and early membrane disruption with minimal inflammatory infiltration.^{27,57} Over time, this evolves into well-defined fibrotic replacement while maintaining underlying tissue architecture, including preserved vascular and neural integrity.⁴² In contrast to RFA, which induces coagulative necrosis and collagen denaturation with collateral tissue injury, PFA results in selective myocyte injury while preserving structural

scaffolding.⁴¹ These features provide the mechanistic basis for the favorable safety profile of PFA and may facilitate more accurate intraprocedural assessment of lesion completeness due to minimal edema and inflammatory response.^{9,58}

Macroscopically, PFA produces broad, contiguous antral lesions encompassing a larger isolated surface area than CBA, which may contribute to improved long-term rhythm control.⁵⁹ Lesion extent and uniformity depend on effective electrode–tissue apposition and consistent electric field distribution.⁶⁰ The integration of impedance-based proximity and real-time electrogram feedback into newer PFA systems represents an important advance toward ensuring reliable lesion delivery and reproducibility.⁵⁹

Beyond tissue-level characteristics, lesion geometry and durability are governed primarily by electric field distribution and catheter–tissue interaction.

DETERMINANTS OF LESION GEOMETRY AND ROLE OF CONTACT FORCE

The determinants of lesion geometry in PFA differ fundamentally from thermal modalities, as lesion formation is governed primarily by electric field distribution rather than conductive heat transfer. Nevertheless, catheter–tissue contact remains an important modulator of effective energy delivery. Adequate contact ensures consistent field coupling and minimizes current dispersion, thereby promoting lesion depth and continuity, whereas poor apposition may result in heterogeneous or subthreshold electric field exposure.⁶¹ Inadequate contact may result in subthreshold electric field exposure, leading to incomplete or nondurable lesion formation.

Preclinical studies using multielectrode platforms, such as VARIPULSE, have demonstrated that increasing contact force (≈ 15 – 30 g) and repeated energy applications independently augment lesion depth and contiguity, underscoring that contact force remains relevant despite the broader electric field profile of PFA.⁶² Similarly, early clinical experience with focal contact force–sensing PFA catheters supports the feasibility of point-by-point ablation with durable lesion formation confirmed on remapping.

In contrast to RFA, where lesion formation is highly dependent on contact force and duration, PFA exhibits relative insensitivity to small variations in contact due to the spatial distribution of the electric field. However, this does not eliminate the importance of contact, particularly in anatomically complex regions or thicker myocardial substrates where adequate field penetration must be ensured.⁶¹

Emerging technologies have incorporated both direct and indirect surrogates of contact into PFA workflows. Dual-energy platforms have introduced composite

Table 3. Comparative Features of PFA Versus Conventional Thermal and Hybrid Energy Modalities

Feature	PFA	RFA	CBA	Hybrid/dual-energy (eg, PFCA, RF+PFA)
Primary mechanism	Nonthermal irreversible electroporation of cardiomyocytes with preservation of extracellular matrix, vasculature, and most neural structures	Resistive and conductive heating causing coagulative necrosis and collagen denaturation	Cryothermal injury via ice-crystal formation and microvascular damage leading to necrosis	Combination of PFA with thermal energy (ultralow-temp cryo or RF) to widen and deepen lesions
Tissue selectivity	High myocyte selectivity; minimal lasting injury to esophagus, PV media, and phrenic nerve in preclinical and registry data	Nonselective; collateral injury to esophagus, PVs, phrenic nerve, and autonomic ganglia well described	Nonselective; PV stenosis rare but phrenic palsy and esophageal injury remain concerns	Intended to retain PFA selectivity while addressing thick tissue; selectivity may be reduced where thermal contribution is large
Acute PVI success	Typically ≥99% in contemporary trials and registries	High (≈95%–100%) in experienced centers	High (≈95%–100%)	PARALELL PFCA: ≈96% PVI, ≈98% PWI; SmartFIRE dual-energy RF+PFA with high acute PVI
12-mo arrhythmia freedom (paroxysmal AF)	≈70%–80% depending on system and trial; ADVENT and PULSED AF show noninferiority to thermal; some series suggest lower recurrence than RFA/CBA	≈65%–75% in contemporary RCTs/meta-analyses	≈60% to 75%; several meta-analyses show slightly higher recurrence vs PFA	Early data (Sphere-360, PFCA, dual-energy) show ≈75%–88% 1-y freedom in persistent and paroxysmal cohorts but follow-up is limited
Procedure time	Shorter than RFA and often shorter than CBA; reductions of ≈10–15 min vs CBA and similar vs or shorter than RF in meta-analyses	Longest among modalities, especially for extensive substrate modification or linear lesions	Single-shot, but set-up and freeze cycles limit efficiency; typically intermediate	PFCA can achieve PVI+PWI through 1 catheter; early data suggest reasonable times but not yet clearly superior
Need for contact force optimization	Less contact-sensitive but still influenced by CF and proximity; some systems lack CF-sensing and use impedance or EGM surrogates	Strongly CF-dependent; inadequate CF associated with gaps and reconnection	Balloon occlusion and pressure at ostium are critical; indirect contact assessment via contrast and pressure tracings	Focal dual-energy and PFCA platforms often CF-sensing; still subject to RF/cold lesion dependence in linear work
Collateral injury: esophagus	To date, no atrioesophageal fistula in large PFA registries (eg, MANIFEST-17K); minor esophageal erythema/edema occasionally reported	Low but nontrivial risk of ulceration and fistula; requires temperature monitoring and avoidance strategies	Rare but documented esophageal injury; protective strategies similar to RF	PFCA and dual-energy experience limited; early PFCA data show no major esophageal events but sample size small
Collateral injury: phrenic nerve	Mostly transient stunning; durable palsy extremely rare in current registries	Palsy risk in SVC/RA and RVOT work; can be permanent	Recognized complication during RSPV/LSPV freezes; sometimes persistent	PFCA series show transient effects only; larger data sets needed
Other complications	Very low rates of PV stenosis, stroke, and death; early signal of higher tamponade in learning curve with stiff catheters; coronary vasospasm described at CTI and mitral isthmus	Vascular complications, tamponade, PV stenosis, stroke, rare fistula; complication profile well known	Vascular complications, tamponade, phrenic palsy; rare PV stenosis and esophageal injury	Theoretical risk of additive complications from both energy types; early PFCA shows low serious event rate but long-term data pending
Durability of non-PV lesions (PWI, lines)	High for posterior wall; more modest for mitral isthmus and complex lines; redo data show frequent reconnection in thick regions, motivating dual-energy approaches	Effective but at cost of higher collateral risk in posterior LA and near coronaries	PWI and complex lines less commonly performed with CBA alone; often require adjunct RF	Designed to improve linear and deep lesion durability (eg, PFCA for CTI, dual-energy for perimitral lines); long-term evidence still emerging

Side-by-side comparison of mechanism, selectivity, efficacy, safety, and durability across PFA, radiofrequency, cryoballoon, and dual-energy modalities. Comparative values reflect typical findings from contemporary clinical trials and registries; specific outcomes vary by operator, protocol, and patient population. AF indicates atrial fibrillation; CBA, cryoballoon ablation; CF, contact force; CTI, cavotricuspid isthmus; EGM, electrogram; LA, left atrium; LSPV, left superior pulmonary vein; PFA, pulsed field ablation; PFCA, pulsed field plus cryoablation; PV, pulmonary vein; PVI, pulmonary vein isolation; PWI, posterior wall isolation; RA, right atrium; RF, radiofrequency; RFA, radiofrequency ablation; RSPV, right superior pulmonary vein; and SVC, superior vena cava.

dosing metrics, such as the pulsed field index, which integrate contact force and application number to standardize lesion delivery.⁶³ Other systems, including balloon-based platforms, use impedance-based measurements as real-time surrogates of tissue contact, while signal-based metrics such as peak frequency have been proposed as functional indicators of effective coupling.^{30,64}

Importantly, not all platforms provide direct contact force feedback. Systems such as lattice or multielectrode

catheters rely on electrode geometry and field distribution rather than precise contact quantification, which may introduce variability in lesion formation. Collectively, these findings highlight that while PFA reduces dependence on contact force compared with TA, optimal lesion geometry still requires adequate tissue coupling, consistent field distribution, and appropriate energy delivery strategies.⁶¹

Consistent with these mechanistic features, PFA lesion durability appears high, although modest regional

variability exists. Voltage mapping typically demonstrates complete PVI with minimal early reconnection, and redo procedures confirm durable isolation in most veins. However, regions such as the mitral isthmus and thick anterior wall remain challenging, likely due to increased tissue thickness or suboptimal catheter contact.⁶⁵ PFA remains effective in reisolating previously ablated segments, supporting its utility in redo procedures and its ability to achieve homogeneous lesions across diverse atrial substrates.

PULSED FIELD VERSUS THERMAL ENERGY

PFA differs fundamentally from thermal modalities in its nonthermal mechanism and selective tissue response. Rather than redescribing these distinctions, this section

focuses on comparative performance across energy sources (Figure 2).

PULSED FIELD VERSUS RADIOFREQUENCY

The biophysical and histological consequences of PFA and RFA are markedly different. Histologically, PFA lesions mature into organized scar tissue characterized by transmural replacement fibrosis without the surrounding inflammation extending to epicardial fat or arterial injury characteristic of RFA lesions, suggesting a more favorable tissue healing response. These mechanistic differences manifest in unique intracardiac electrogram signatures that may offer opportunities for real-time lesion assessment. Although both modalities cause an immediate reduction in bipolar electrogram

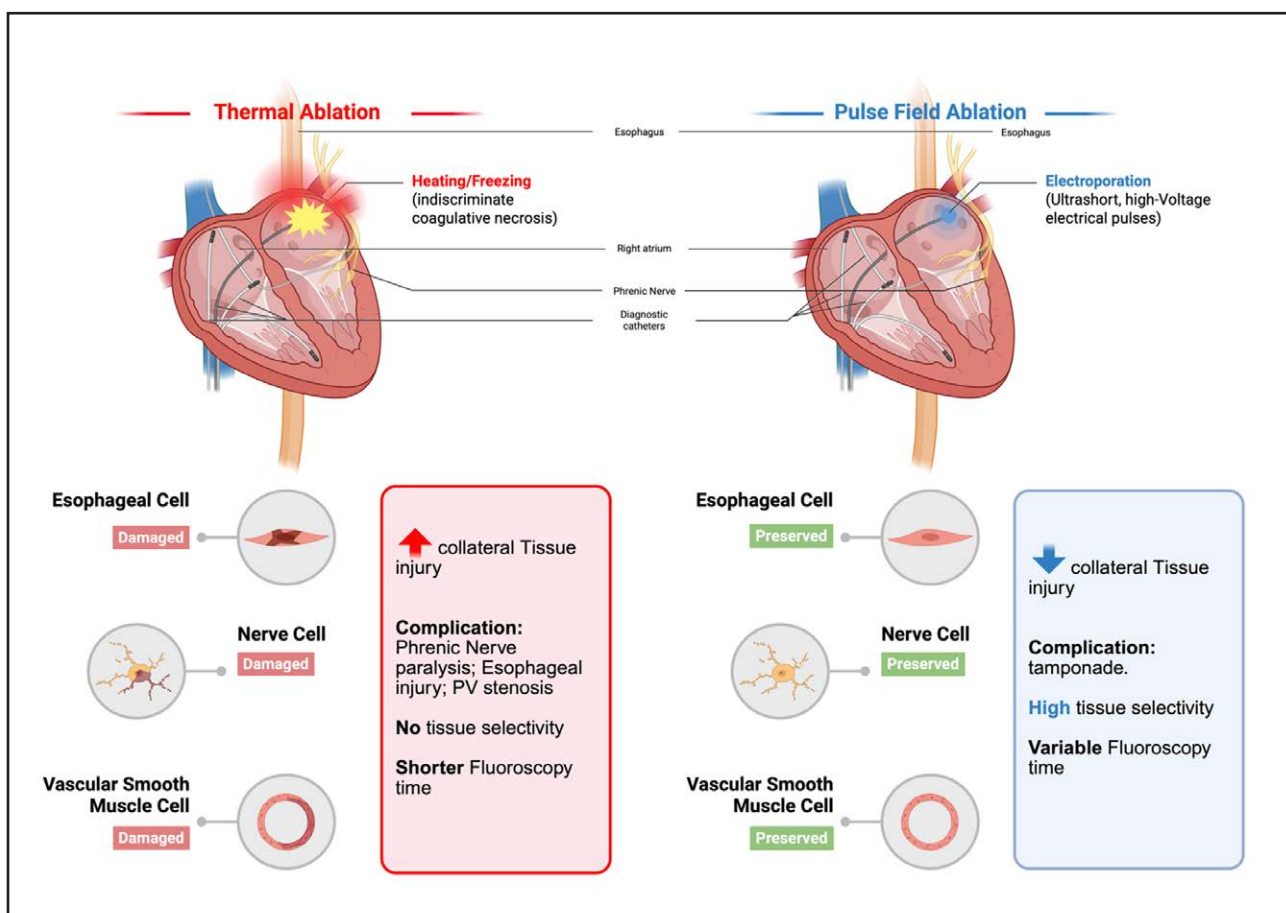


Figure 2. Comparison of thermal ablation and pulse field ablation (PFA) mechanisms and collateral tissue injury.

This figure contrasts the mechanisms of action and the resulting collateral damage profiles of conventional thermal ablation (radiofrequency or cryoablation, **left**) and PFA (**right**). Thermal Ablation relies on heating or freezing to cause indiscriminate coagulative necrosis of myocardial tissue (**top left**). This nonselective energy delivery can lead to significant collateral damage in adjacent nonmyocardial structures (**bottom left**), including (1) esophageal injury, leading to the risk of atrioesophageal fistula; (2) phrenic nerve injury, causing nerve paralysis; and (3) damage to the vascular smooth muscle, potentially leading to pulmonary vein (PV) stenosis. PFA uses ultrashort, high-voltage electrical pulses (electroporation) to achieve tissue destruction (**top right**). PFA demonstrates high tissue selectivity for cardiac myocytes over nonmyocardial cells (**bottom right**), thereby preserving the integrity of the esophageal wall, phrenic nerve, and vascular smooth muscle. The primary acute complication associated with PFA is cardiac tamponade, but it generally offers a reduced risk of collateral tissue injury compared with thermal methods.

amplitude, PFA uniquely induces a pronounced, transient increase in low-frequency (1–16 Hz) unipolar electrogram amplitude—a current of injury phenomenon. In a porcine model, the recovery dynamics of this current of injury differed significantly between transmural and nontransmural PFA lesions as early as 60 seconds postablation ($P < 0.001$), a distinction not observed with RFA. However, a critical caveat of PFA is that the immediate disappearance of bipolar electrograms can result from transient cellular stunning due to reversible electroporation, and does not guarantee a durable, permanent lesion. Nontransmural PFA lesions have demonstrated significant bipolar electrogram recovery within minutes, with studies showing that electrograms recovered in regions with reversible effect within 5 minutes, a phenomenon not seen with RFA. These findings underscore that traditional electrophysiological end points developed for RFA may not be directly applicable to PFA, necessitating the development of modality-specific criteria for durable lesion assessment.⁶⁶

PULSED FIELD VERSUS CRYOABLATION

When compared with CBA, PFA demonstrates significant advantages in procedural efficiency and safety. PFA achieves a markedly higher first-pass isolation rate of 98.8% for PFA versus 81.5% for cryoballoon and 73.1% for radiofrequency ($P < 0.001$).⁶⁷ This enhanced efficiency is likely attributable to the mechanism of PFA, which is less dependent on perfect, continuous tissue contact than the cryothermal energy delivery of a balloon catheter.

The safety advantages of PFA are particularly pronounced in comparison to CBA. The risk of phrenic nerve injury, a known complication of CBA, is substantially lower with PFA (odds ratio, 0.21 [95% CI, 0.09–0.49]). However, early clinical experience with PFA was associated with a higher risk of cardiac perforation or tamponade compared with TA (odds ratio, 2.98 [95% CI, 1.27–7.00]).⁶⁸ This risk appears to be related to the learning curve and technical factors, such as the stiffness of early generation catheters and guidewires, as complication rates have declined with increased operator experience and equipment refinement. Although reconnection patterns between different ablation modalities have been studied, the lowest reconnection rates were observed for the RIPVs across technologies, though specific comparative rates between PFA and TA require further investigation. These data position PFA as a highly effective and efficient alternative to TA, offering a potentially superior long-term efficacy and safety profile, although careful attention to procedural technique remains paramount to mitigate mechanical risks.

TISSUE SELECTIVITY

Extensive preclinical and clinical studies have shown that PFA selectively affects heart muscle cells while preserving the surrounding connective tissue, blood vessels, and nerves.^{1,10,23,59,69} This tissue-specific effect greatly reduces the risk of complications, such as esophageal injury, PV narrowing, and phrenic nerve injury, providing a safety profile that is superior to traditional TA methods.^{16,59,69}

PV ISOLATION

PFA achieves high acute and durable PVI. In pooled analyses of early trials, acute success exceeded 99% and durable isolation reached 96% with optimized biphasic waveforms.⁷⁰ Large registries, such as MANIFEST-PF, confirmed excellent safety with minimal esophageal or phrenic nerve injury. These results collectively validate PFA as an effective and durable strategy for PVI.⁷¹

NON-PV SUBSTRATES



Although PVI remains the cornerstone of AF therapy, durable control of persistent AF often requires ablation of non-PV triggers and substrate modification. The favorable safety profile of PFA makes it particularly well-suited for anatomically vulnerable regions.

Posterior Wall

The posterior left atrial wall is a frequent substrate for AF but is challenging to ablate with thermal energy due to esophageal proximity. PFA has demonstrated high efficacy for posterior wall isolation, typically achieved with a box lesion set connecting the PVs. In ADVANTAGE-AF, acute posterior wall isolation success reached 99.7% without esophageal complications, and durability was confirmed in ≈85% of redo procedures with only minor lesion regression.⁷² These findings suggest that PFA may mitigate one of the greatest limitations of TA—risk of atrioesophageal fistula—while achieving durable posterior wall block.

Ligament and Vein of Marshall

The ligament of Marshall (containing the vein of Marshall) is a recognized arrhythmogenic source and contributor to perimitral flutter. Ethanol infusion into the vein of Marshall has been the established isolation technique, with success rates of 88.9% but risks of vessel dissection and coronary spasm.⁷³ Although PFA has proven effective for complex atrial substrates, its specific role in Marshall bundle ablation remains largely unexplored.

Given its tissue selectivity, PFA could offer advantages in this autonomic-rich region adjacent to the coronary sinus and left atrial appendage (LAA), but clinical data are still lacking. Early preclinical data suggest that PFA is not durable for vein of Marshall ablation and still requires additional ethanol ablation. Often the area of the left atrium where the vein of Marshall runs along has variable thickness, and a deeper predictable lesion may be harder to achieve.

Left Atrial Appendage

LAA isolation may benefit selected persistent AF patients but carries a high risk with radiofrequency due to thin walls and adjacent critical structures. The safety of PFA makes it an attractive alternative. Early hybrid approaches combining PFA-based isolation with mechanical LAA occlusion have demonstrated feasibility: in canine studies, acute success was 100% with 80% durability and complete endothelialization at follow-up, without device-related complications.^{74,75} In patients with preexisting LAA closure devices, PFA could reduce perforation risk and preserve mechanical function, although data are limited and require further evaluation. The left circumflex runs in close proximity to the LAA ostium and can be at risk for coronary vasospasm induced by PFA.

Perimitral Isthmus

Achieving durable conduction block along the mitral isthmus is notoriously difficult with thermal energy. PFA has shown high acute success in creating bidirectional block, yet long-term durability remains suboptimal, with frequent reconnection on redo procedures.⁷⁶ Hybrid catheter systems capable of toggling between PFA and RFA have emerged, potentially combining the rapid, safe lesion creation of PFA with the greater depth of RFA for this challenging site. Predictors of successful PFA for perimitral lines include shorter left atrial dwell times and lack of prior antiarrhythmic drug failure.⁷⁷ One has to pay attention to the course of the left circumflex coronary artery and associated risk of vasospasm in this region.

Cavotricuspid Isthmus

The cavotricuspid isthmus is the standard target for typical atrial flutter ablation. PFA has been highly effective in producing acute, durable bidirectional block.^{78,79} Rare but serious cases of coronary vasospasm have been described during cavotricuspid isthmus ablation, highlighting the need for procedural vigilance.^{80,81} A novel hybrid approach—pulsed field cryoablation—combines PFA with ultralow temperature cryoablation to generate wider, transmural cavotricuspid isthmus lesions, eliminating PFA-induced muscle contraction and microbubble formation in preclinical studies.⁸²

Superior and Inferior Vena Cava

The superior vena cava is a well-recognized non-PV trigger and may be inadvertently affected during PFA of the right superior PV. Clinical studies have documented transient superior vena cava stunning or isolation, particularly in anatomies with close right superior PV—superior vena cava proximity.^{83,84} Although typically benign and reversible, this underscores the potent biological effects of PFA. The inferior vena cava has also been studied in preclinical models as a safety landmark: direct PFA delivery to the inferior vena cava, with the esophagus displaced against it, produced no esophageal injury, reinforcing the remarkable tissue selectivity of PFA.¹ Often, the right phrenic nerve runs in close proximity to the lateral aspects of the superior vena cava, crista terminalis, and inferior vena cava and is at risk of injury with RFA or CBA. Theoretically, this issue may be significantly reduced with PFA, although there is emerging data suggesting the acute stunning effect on the Phrenic nerve.⁸⁵

CATHETERS AND TOOLS

Catheter design in PFA is fundamentally centered on achieving predictable electric field distribution while maintaining tissue selectivity and procedural efficiency. Unlike TA systems, where lesion formation is governed by conductive heat transfer, PFA catheter architecture directly determines electric field geometry, depth of penetration, and spatial uniformity. As a result, electrode configuration, spacing, and energy delivery mode (bipolar versus monopolar) serve as the primary engineering determinants of lesion formation.

Building on these principles, the rapid evolution of PFA has been paralleled by the development of a diverse array of catheter designs optimized for different lesion delivery strategies. These technologies are broadly categorized by their design philosophy: large-area, single-shot catheters engineered for efficient circumferential PVI, and focal or intermediate-footprint catheters designed for precise, point-by-point ablation of both PV and non-PV targets. A comparative overview of currently available and investigational PFA catheter systems, including their design characteristics and energy delivery strategies, is provided in Figure 3.

Key engineering considerations include electrode configuration (eg, multispline, lattice, or focal designs), electrode spacing, and bipolar versus monopolar energy delivery, all of which influence field uniformity and lesion geometry. Contact characteristics also play a critical role; while some systems rely on tissue contact for effective energy coupling, others are designed to deliver more diffuse, noncontact fields. In addition, integration with electroanatomical mapping systems and the ability to combine mapping and ablation functions have become important design features in newer platforms. These

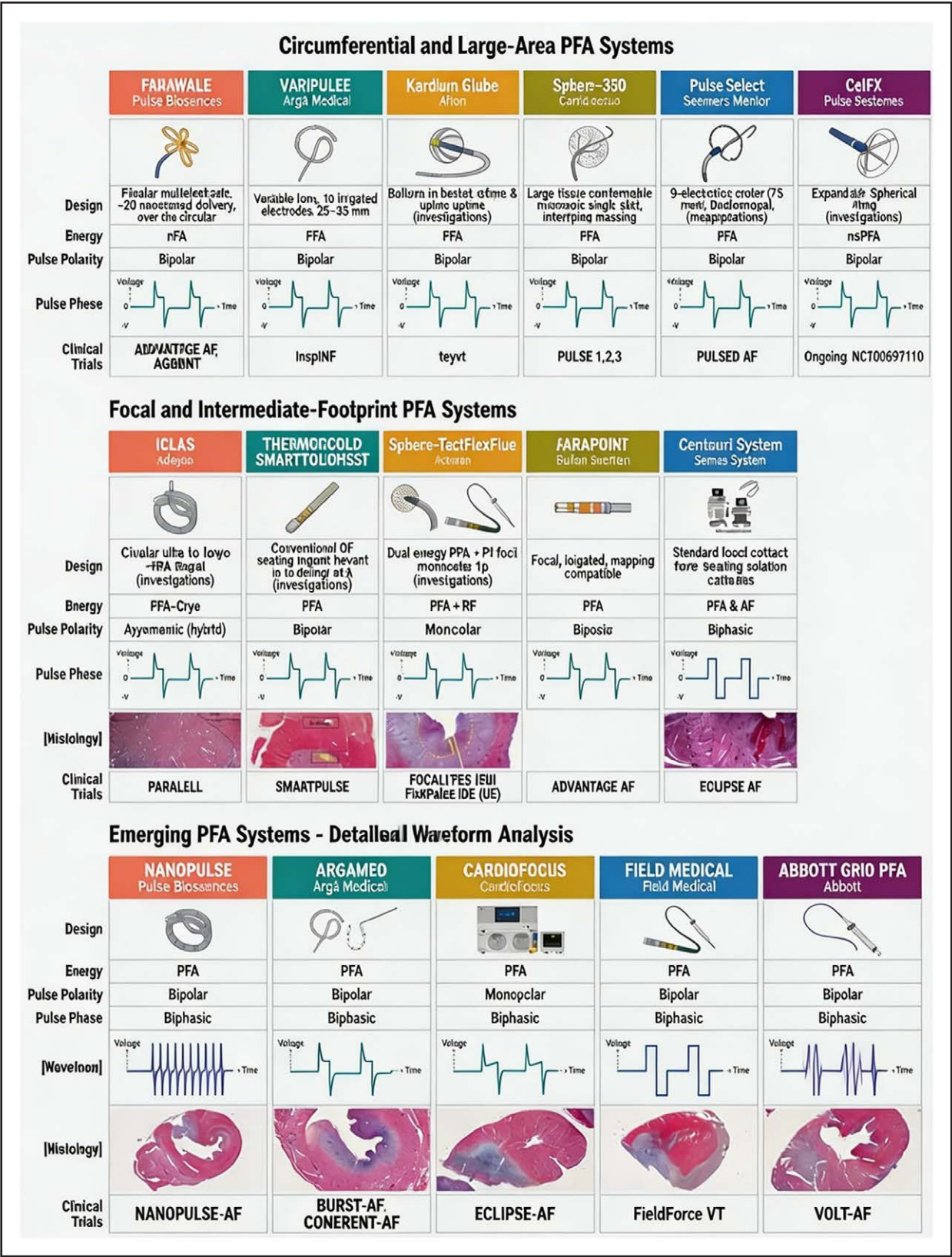


Figure 3. Pulsed field ablation (PFA) systems and evolution of waveform design. Schematic overview of contemporary and emerging PFA platforms, grouped by ablation footprint and catheter architecture. The figure highlights key differences in energy delivery, pulse polarity, and waveform configuration across systems, with representative lesion histology and associated clinical trial experience. The transition from early monophasic to optimized biphasic and multipulse waveforms underscores ongoing efforts to improve lesion durability while reducing off-target effects and procedural variability. AF indicates atrial fibrillation; CF, contact force; Cryo, cryoablation; PV, pulmonary vein; and RF, radiofrequency.

engineering parameters collectively determine the balance between lesion durability, procedural efficiency, and risk of off-target effects, forming the basis for the diversity of contemporary PFA catheter platforms.

Contemporary PFA systems can be broadly categorized based on catheter architecture and energy delivery strategy. Balloon-spline and multispline systems (eg, Farapulse) are designed for circumferential PVI with rapid, wide-area lesion sets. Lattice and focal catheters (eg, Affera Sphere-9) enable more flexible lesion delivery, including linear and substrate-based ablation, often with integrated mapping capabilities. Other investigational platforms use variable electrode arrays or hybrid energy modalities to optimize lesion depth and field control. These design differences reflect trade-offs between procedural efficiency, lesion precision, and adaptability to complex atrial substrates.

CIRCUMFERENTIAL AND LARGE-AREA PFA SYSTEMS

These systems are designed to simplify the PVI procedure by delivering energy to a wide tissue area with a limited number of applications, thereby reducing cardiac chamber dwell times and operator variability.

FARAWAVE (Boston Scientific)

The FARAWAVE catheter (Boston Scientific, Marlborough, MA) is a pentaspline, flower-shaped multielectrode system designed to achieve circumferential PV coverage. Its expandable spline architecture with distributed electrodes enables simultaneous energy delivery across a wide antral surface, promoting uniform electric field distribution and rapid lesion creation. This single-shot design reduces procedural time and operator variability, making it well-suited for PVI and other large-area ablation targets. However, its large footprint and extensive field coverage may result in greater myocardial exposure, reflected by higher biomarker elevations and potential hemolysis, highlighting the importance of optimized energy dosing and procedural monitoring.⁸⁷

MEDTRONIC PULSESELECT

PulseSelect (Medtronic)

The PulseSelect system (Medtronic, Minneapolis, MN) is a circular multielectrode catheter delivering biphasic monopolar pulses. Its fixed-loop configuration is designed to provide consistent circumferential contact and controlled electric field distribution at the PV ostium. The monopolar delivery strategy enables broader field penetration and efficient lesion formation. However, field distribution may be more sensitive to anatomic variation and catheter positioning, and transient electromagnetic

interactions with cardiac implantable devices have been reported.⁸²

Volt (Abbott)

The Volt system (Abbott, Chicago, IL) utilizes a balloon-in-basket configuration integrated with electroanatomic mapping. This hybrid design enables circumferential tissue contact while maintaining positional stability, facilitating relatively uniform electric field distribution across variable PV anatomies. Integration with mapping systems supports real-time lesion assessment and procedural efficiency. However, lesion quality remains dependent on balloon apposition and anatomic conformity, which may vary across patients.⁵⁸

Sphere-360 (Medtronic)

The Sphere-360 catheter (Medtronic) is a large lattice-based, single-shot system designed to integrate high-density mapping and ablation. Its spherical lattice configuration enables broad electric field coverage and rapid PVI with minimal catheter manipulation. Iterative waveform optimization further enhances field uniformity and lesion durability. However, effective lesion formation depends on consistent tissue engagement across the lattice, and excessive field exposure may increase myocardial injury if not carefully controlled.^{33,71}

Kardium Globe

The Globe Mapping and Ablation System (Kardium, Vancouver, Canada) employs a spherical multielectrode basket with a high-density electrode array designed for simultaneous mapping and PFA delivery. This configuration enables comprehensive electric field coverage and rapid identification of arrhythmogenic substrates while delivering circumferential lesions.^{88,89} The integrated mapping-ablation capability may enhance procedural efficiency in complex atrial substrates. However, its large device profile and reliance on uniform electrode contact may introduce variability in field distribution across heterogeneous anatomy.⁹⁰

FOCAL AND INTERMEDIATE-FOOTPRINT PFA SYSTEMS

These catheters are designed for point-by-point energy delivery, offering the precision required for creating linear lesions, ablating non-PV triggers, or performing touch-up ablation after a primary PVI.

VARIPULSE (Biosense Webster)

The VARIPULSE system (Biosense Webster, Irvine, CA) utilizes a variable-loop circular catheter delivering bipolar biphasic pulses and is integrated with the CARTO 3

mapping system.^{91,92} Its adjustable loop design allows adaptation to PV anatomy, improving electrode–tissue contact and electric field coupling. Bipolar energy delivery enables localized field confinement, enhancing safety and reducing off-target effects. However, lesion formation remains dependent on optimal catheter positioning and may require repeated applications in complex anatomic regions.^{93,94}

FARAPOINT (Boston Scientific)

The FARAPOINT catheter (Boston Scientific) is a focal, irrigated PFA catheter designed for precise, point-by-point lesion delivery. Its compact electrode configuration enables targeted ablation of non-PV triggers and facilitates touch-up applications after circumferential ablation. Compatibility with the FARAWAVE system allows seamless transition between wide-area and focal strategies within a single procedure. However, focal delivery requires greater procedural time and operator precision, and lesion continuity depends on meticulous catheter manipulation.⁹⁵

Sphere-9 (Medtronic)

The Sphere-9 catheter (Medtronic) is a focal, irrigated lattice-tip catheter featuring a 9-mm spherical electrode array designed for dual-energy delivery using both pulsed field and radiofrequency energy. This design provides flexibility for creating focal and linear lesions, particularly in complex substrates. The lattice geometry enables controlled electric field distribution while maintaining stable tissue contact. However, integration of dual-energy modalities introduces additional complexity in dosing and procedural workflow.⁹⁶

Emerging and Niche Technologies

These systems represent innovative or regionally developed approaches that extend beyond mainstream PFA platforms. They explore new concepts such as hybrid energy delivery, ultrashort nanosecond pulse technology, or focal contact force–guided ablation, with early studies suggesting promising safety and efficacy profiles.

CellFX (Pulse Biosciences, nsPFA)

The CellFX system (Pulse Biosciences, Hayward, CA) delivers ultrashort nanosecond-duration pulses designed to target intracellular structures and induce apoptosis through high-intensity electric fields. This approach enables highly nonthermal energy delivery with minimal conductive heating and potential preservation of surrounding tissue integrity. However, the requirement for higher peak electric fields and limited long-term clinical data remain important considerations.⁹⁷

Adagio iCLAS (Pulsed Field Cryoablation)

The iCLAS system (Adagio Medical, Laguna Hills, CA) combines PFA with ultralow temperature cryoablation, creating a hybrid pulsed field cryoablation approach. Precooling the tissue increases impedance and concentrates the electric field, allowing deeper and more uniform lesion formation while minimizing microbubble formation. However, the integration of dual-energy modalities introduces complexity in energy delivery and procedural control.⁹⁸

Centauri Platform

The Centauri platform delivers biphasic monopolar pulsed electric fields through standard contact force–sensing focal catheters integrated with existing mapping systems. This design leverages familiar catheter platforms while enabling pulsed field energy delivery, facilitating workflow integration. The monopolar configuration allows broader field penetration but requires careful waveform optimization to limit off-target effects.⁶²



Nanopulse (Pulse Biosciences)

The Nanopulse system employs high-repetition-frequency nanosecond pulses delivered in a bipolar configuration. These ultrashort pulses achieve electroporation thresholds with minimal thermal effect, while rapid repetition enables cumulative electroporation and transmural lesion formation. However, higher peak electric fields are required, and parameter optimization remains an area of ongoing investigation.

Argamed

The Argamed platform utilizes customizable biphasic or asymmetrical biphasic waveforms with adjustable voltage, pulse width, and frequency. This flexibility enables fine-tuning of electric field distribution and lesion depth while reducing neuromuscular stimulation. However, variability in waveform definitions and parameter selection may limit standardization and cross-platform comparability.

CardioFocus Centauri (WAVE1)

The CardioFocus Centauri system delivers a custom monopolar biphasic waveform through standard contact force–sensing focal catheters. Energy is applied between the catheter tip and a dispersive patch, generating a broader and deeper electric field compared with bipolar systems. Although this configuration provides flexibility for focal and linear ablation, it requires precise waveform control to mitigate skeletal muscle activation and off-target effects.

Field Medical

The Field Medical system employs proprietary biphasic waveforms delivered through bipolar catheter configurations designed to confine the electric field to the immediate vicinity of the electrode. This localized delivery reduces off-target exposure and enhances procedural safety. However, fixed dosing protocols may limit adaptability across different anatomic substrates.

Volt Grid System (Abbott)

The Volt grid-based PFA system utilizes a multielectrode lattice configuration to distribute electric fields across a broad surface area. This design promotes uniform lesion formation even in the presence of variable tissue contact. However, high-current delivery and non-R-wave-gated application require careful waveform optimization to minimize arrhythmogenic risk and skeletal muscle activation.

SUMMARY

Early experiences with nanosecond PFA, pulsed field cryoablation, and contact force-guided focal PFA consistently demonstrate high acute success and acceptable safety in preliminary studies. Long-term data are awaited to clarify whether these approaches improve lesion durability or expand indications beyond AF.

CONCLUSIONS

PFA represents a pivotal advancement in catheter ablation, offering rapid, tissue-selective lesion creation with a strong safety record. Future development should focus on optimizing waveform delivery, integrating real-time lesion assessment, and expanding use to complex atrial and ventricular substrates. The second section will summarize clinical findings and practical approaches for integrating PFA into regular electrophysiology practice.

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